

Supplementary Table S7. Hi-C Feature Overlap Statistics for 21 lncRNA-DMB Pairs

Data source: GSE186599 (Johnston et al., Cell, 2024). Background overlap rate: 10% (genome-wide random window sampling). One-sided exact binomial test; Bonferroni-corrected for 5 simultaneous tests ($\alpha = 0.05/5 = 0.010$).

Significance: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; n.s. not significant.

Part 1 — Individual lncRNA-DMB pairs (n = 21)

DMB pair ID	lncRNA	Chr location	Gene coordinates (hg38)	Functional class	DMB index	Mean $ \Delta\beta $	Dir. consist. (%)	Priority score	Diff. loops (HiCCUPS)	TAD boundaries	Pan-tumour TULIPs	PFA-specific TULIPs	Ins. score minima (all)	No. Hi-C features (of 5)	Hi-C support tier
DMB-01	TARID	9q22.32	chr9:101,416,400-101,431,200	Demethylation regulator	1	0.357	100%	7.28	✓	✓	✓	✓	✓	5	High (≥4/5)
DMB-02	TARID	9q22.32	chr9:101,416,400-101,431,200	Demethylation regulator	2	0.357	100%	7.28	✓	✓	✓	✓	✓	5	High (≥4/5)
DMB-03	TARID	9q22.32	chr9:101,416,400-101,431,200	Demethylation regulator	3	0.357	100%	7.28	✓	✓	–	–	✓	3	High (≥4/5)
DMB-04	TARID	9q22.32	chr9:101,416,400-101,431,200	Demethylation regulator	4	0.357	100%	7.28	✓	–	–	–	–	1	High (≥4/5)
DMB-05	TARID	9q22.32	chr9:101,416,400-101,431,200	Demethylation regulator	5	0.357	100%	7.28	✓	–	–	–	–	1	High (≥4/5)
DMB-06	DINO	2q31.1	chr2:172,810,000-172,818,500	DNA damage / p53 amplifier	1	0.264	100%	5.82	✓	✓	✓	✓	✓	5	High (≥4/5)
DMB-07	DINO	2q31.1	chr2:172,810,000-172,818,500	DNA damage / p53 amplifier	2	0.264	100%	5.82	✓	✓	–	–	✓	3	High (≥4/5)
DMB-08	DINO	2q31.1	chr2:172,810,000-172,818,500	DNA damage / p53 amplifier	3	0.264	100%	5.82	–	–	–	–	–	0	None (0/5)
DMB-09	H19	11p15.5	chr11:1,995,000-2,004,000	Imprinted / ceRNA hub	1	0.231	75%	5.18	✓	✓	✓	–	✓	4	High (≥4/5)
DMB-10	H19	11p15.5	chr11:1,995,000-2,004,000	Imprinted / ceRNA hub	2	0.231	75%	5.18	✓	✓	–	–	✓	3	High (≥4/5)
DMB-11	H19	11p15.5	chr11:1,995,000-2,004,000	Imprinted / ceRNA hub	3	0.231	75%	5.18	✓	–	–	–	–	1	High (≥4/5)
DMB-12	H19	11p15.5	chr11:1,995,000-2,004,000	Imprinted / ceRNA hub	4	0.231	75%	5.18	–	–	–	–	–	0	None (0/5)
DMB-13	MALAT1	11q13.1	chr11:65,497,000-65,507,000	Nuclear arch. / splicing	1	0.198	83%	5.15	✓	✓	–	–	✓	3	High (≥4/5)
DMB-14	MALAT1	11q13.1	chr11:65,497,000-65,507,000	Nuclear arch. / splicing	2	0.198	83%	5.15	✓	–	–	–	–	1	High (≥4/5)
DMB-15	MALAT1	11q13.1	chr11:65,497,000-65,507,000	Nuclear arch. / splicing	3	0.198	83%	5.15	–	–	–	–	–	0	None (0/5)

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DMB-16	KCNQ1OT1	11p15.5	chr11:2,608,000-2,679,000	Imprinted / silencing	1	0.187	100%	4.92	✓	✓	–	–	✓	3	High (≥4/5)
DMB-17	KCNQ1OT1	11p15.5	chr11:2,608,000-2,679,000	Imprinted / silencing	2	0.187	100%	4.92	–	–	–	–	–	0	None (0/5)
DMB-18	LINC00261	20p11.21	chr20:22,680,000-22,693,000	TSG context	1	0.173	50%	4.21	–	–	–	–	–	0	None (0/5)
DMB-19	LINC00261	20p11.21	chr20:22,680,000-22,693,000	TSG context	2	0.173	50%	4.21	–	–	–	–	–	0	None (0/5)
DMB-20	LINC00261	20p11.21	chr20:22,680,000-22,693,000	TSG context	3	0.173	50%	4.21	–	–	–	–	–	0	None (0/5)
DMB-21	LINC00261	20p11.21	chr20:22,680,000-22,693,000	TSG context	4	0.173	50%	4.21	–	–	–	–	–	0	None (0/5)

Supplementary Table S7 (continued) — Part 2: Statistical Enrichment Analysis

One-sided exact binomial test ($H_0: p_{obs} \leq 0.10; H_1: p_{obs} > 0.10$). Background rate = 10% treated as known population proportion. Bonferroni correction: $\alpha = 0.05/5 = 0.010$. Method: `scipy.stats.binomtest` (Python 3.12, `scipy` 1.13).

Hi-C Feature	k observed (/ 21 pairs)	Observed rate	Background rate	Fold enrichment	p-value (raw, one-sided)	p-value (Bonferroni ×5)	Significance	Reject H0?
Differential chromatin loops (HiCCUPS)	13/21	61.9%	10%	6.19x	$p = 9.3 \times 10^{-9}$	$p = 4.7 \times 10^{-8}$	***	Yes
TAD boundaries (insulation score minima)	9/21	42.9%	10%	4.29x	$p = 9.5 \times 10^{-5}$	$p = 4.8 \times 10^{-4}$	***	Yes
Pan-tumour TULIPs	4/21	19.0%	10%	1.90x	$p = 1.5 \times 10^{-1}$	$p = 7.6 \times 10^{-1}$	n.s.	No
PFA-specific TULIPs	3/21	14.3%	10%	1.43x	$p = 3.5 \times 10^{-1}$	$p = 1.0 \times 10^0$	n.s.	No
Insulation score minima (all boundaries)	9/21	42.9%	10%	4.29x	$p = 9.5 \times 10^{-5}$	$p = 4.8 \times 10^{-4}$	***	Yes

Notes

- Hi-C features from GEO GSE186599 (Johnston et al., Cell 187:1174-1191.e23, 2024); 140 paediatric brain tumour samples; 10 kb resolution.
- Overlap defined as lncRNA-DMB interval (± 100 kb around gene locus) within 25 kb of any Hi-C feature anchor or boundary on the same chromosome.
- Background rate (10%) estimated by random sampling of 10,000 genomic windows matched for CpG density and chromosomal distribution.
- Exact binomial test preferred over Fisher's exact test because background rate is a known population proportion derived from genome-wide sampling, not an observed count.
- All lncRNA-DMB associations are computational predictions requiring experimental validation in MB cell models (e.g. TARID overexpression, dCas9-TET2 demethylation, miR-181a-5p inhibition).